

Required practical: I–V characteristics

Year 10 GCSE Physics, AQA Trilogy RP 16; Physics RP 4

Name _____

Class _____ Date _____

Plan. Measure. Explain.

Work in pairs. Read the whole method before you build anything.

Online version with hints: panphy.app/gcsephy/year10phy/unit01/practical/iv-characteristics/

1 Aim

Use circuit diagrams to construct circuits that investigate the I–V characteristics of a filament lamp, a diode and a resistor at constant temperature.

$$R = V \div I$$

R in ohms (Ω), V in volts (V), I in amperes (A)

2 Apparatus

- Variable d.c. power supply (0–6 V) or battery pack
- Variable resistor (rheostat)
- Ammeter (0–1 A) and milliammeter (0–100 mA), or multimeters
- Voltmeter (0–10 V)
- Switch and connecting leads
- Fixed resistor, for example 10 Ω
- Filament lamp, for example 6 V 0.3 A
- Diode and a protective resistor, for example 100 Ω

3 Safety

- The lamp gets hot. Do not touch it while it is on or just after.
- Do not go above the lamp's rated p.d. (for example 6 V).
- Always use the protective resistor with the diode, so the current stays small.

Before you start

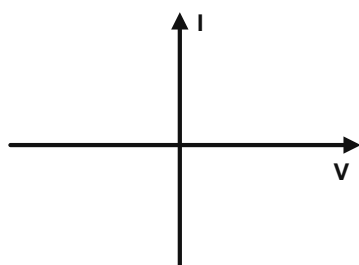
Learn the key words, then make a prediction before you build anything.

4 Key words

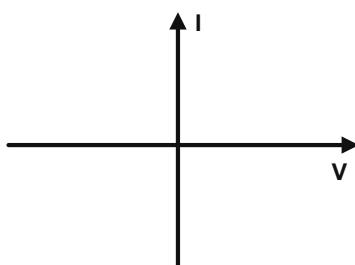
Independent variable	The variable you change.
Dependent variable	The variable you measure.
Control variable	A variable you keep the same, so the test is fair.
Resolution	The smallest change a meter can show, for example 0.01 A.
Anomalous result	A result that does not fit the pattern of the others.
I–V characteristic	A graph of current (y-axis) against p.d. (x-axis) for a component.
Ohmic conductor	A component whose current is directly proportional to the p.d. at constant temperature.

5 Predict

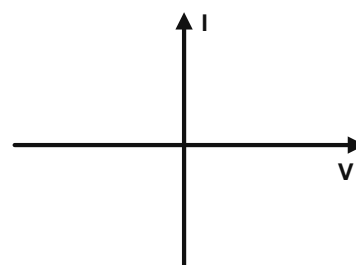
Sketch the I–V graph you expect for each component. Show both positive and negative p.d.



Fixed resistor



Filament lamp



Diode

Method Set up and measure

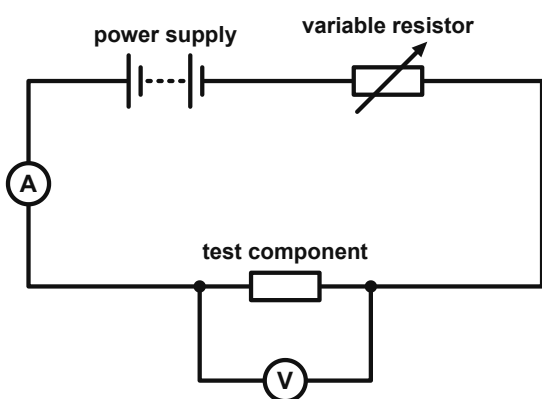
I–V characteristics, Method: Use the variable resistor to change the p.d. across each component. Record the current each time, then reverse the supply and repeat.

1 Variables

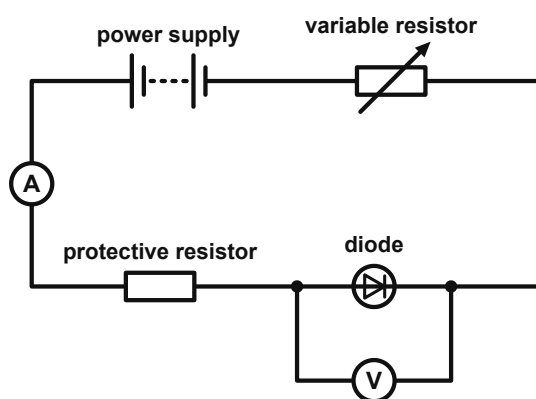
Variable	What it is	How
Independent	p.d. across the component	Adjust the variable resistor or the supply
Dependent	Current through the component	Read the ammeter
Control	Temperature of the fixed resistor	Small currents; switch off between readings

2 Method

1. Build the first circuit below with the fixed resistor as the test component. The voltmeter goes across the component only. ☐
2. Set the variable resistor to its highest resistance. Close the switch and record the p.d. and the current. ☐
3. Change the variable resistor to increase the p.d. in steps of about 1 V. Record the p.d. and current each time. Switch off between readings so the resistor stays cool. ☐
4. Swap the connections to the power supply so the current flows the other way. Repeat steps 2 and 3. Record these p.d. and current values as negative numbers. ☐
5. Replace the resistor with the filament lamp and repeat steps 2 to 4. Do not go above the lamp's rated p.d. ☐
6. Build the second circuit for the diode. Use a milliammeter and increase the p.d. in small steps of about 0.1 V from 0.4 V, where the current starts to rise quickly. Then reverse the supply and take readings. ☐



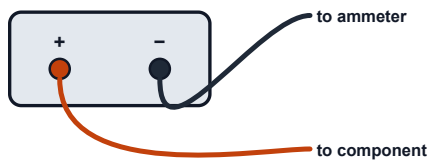
Test the resistor first, then swap in the filament lamp.



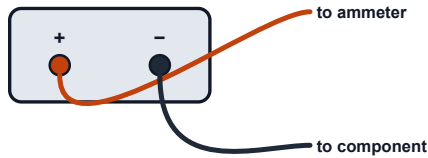
For the diode: add a protective resistor and use a milliammeter. The voltmeter goes across the diode only.

Method Set up and measure

Forward: record + values



Reversed: swap the two leads



Swap the two leads at the power supply. The meters now show negative readings: write the minus sign.

real diode



symbol



It conducts when the current flows towards the band (the bar).

The band on the diode matches the bar in its symbol.
Forward: current flows towards the band.

Why reverse the supply? An I–V graph shows both directions. Reversing the connections gives the negative half of the graph.

Using the variable resistor. Start at its highest resistance, so the p.d. across the component is small. Lower its resistance a step at a time to increase the p.d.

Results Record your readings

I–V characteristics, Results: Record every reading as you take it. Reversed readings are negative: write the minus sign.

1 Results: fixed resistor

Forward		Reversed (negative)	
p.d. / V	Current / A	p.d. / V	Current / A

2 Results: filament lamp

Calculate the resistance, $R = V \div I$, for each forward reading.

Forward			Reversed (negative)	
p.d. / V	Current / A	Resistance / Ω	p.d. / V	Current / A

Results Record your readings

3 Results: diode

Use a milliammeter. Take small steps near 0.6 V.

Forward		Reversed (negative)	
p.d. / V	Current / mA	p.d. / V	Current / mA

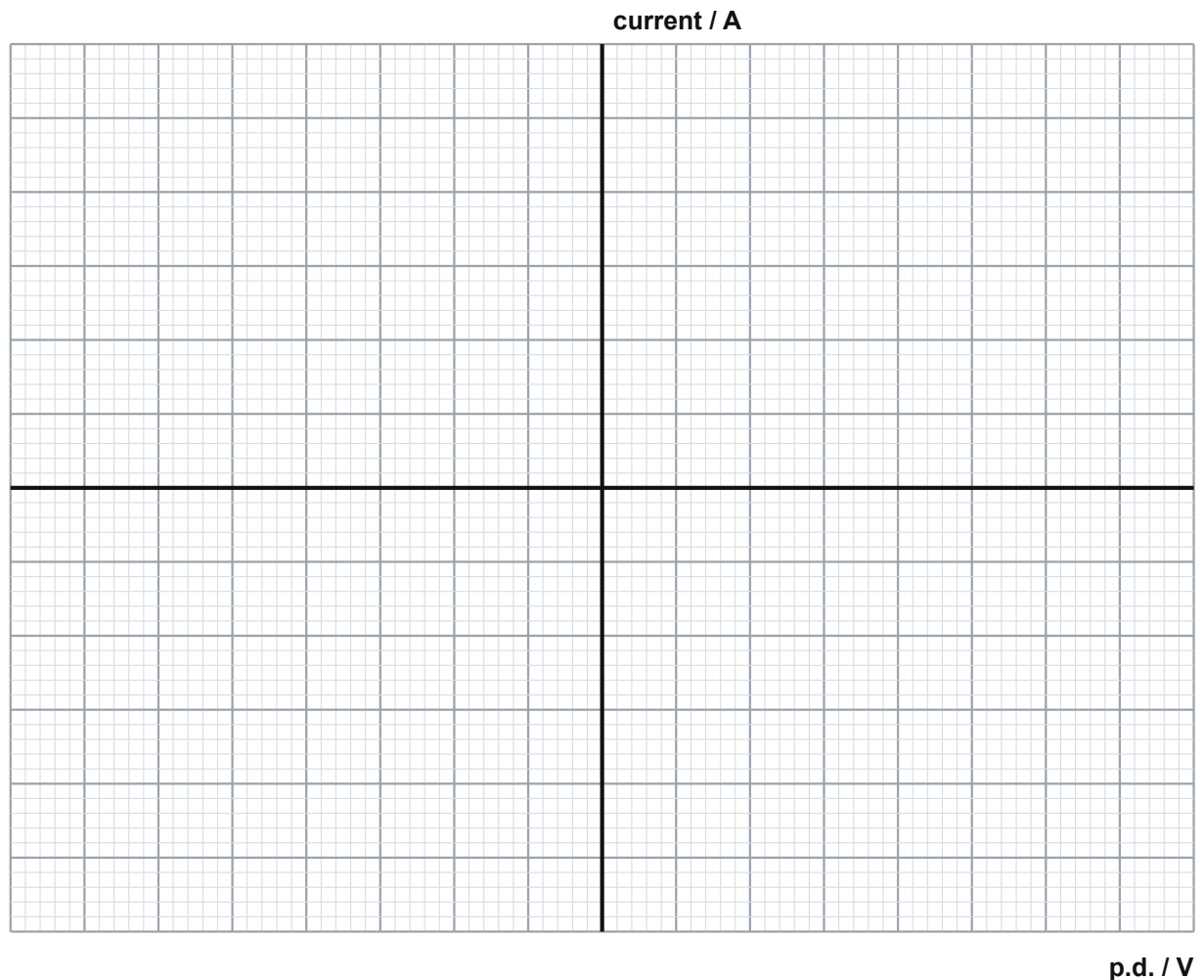
Check your readings. Every graph should pass through the origin: with no p.d. across a component, there is no current through it.

Analysis Graphs and conclusions

I–V characteristics, Analysis: Plot current (y-axis) against p.d. (x-axis) for each component, using the positive and negative readings. Then explain each shape.

1 Fixed resistor graph

Plot current against p.d. for the fixed resistor. Add scales to both axes. Draw a line of best fit.



Q1 Describe the I–V graph for the fixed resistor. What does it tell you about its resistance?

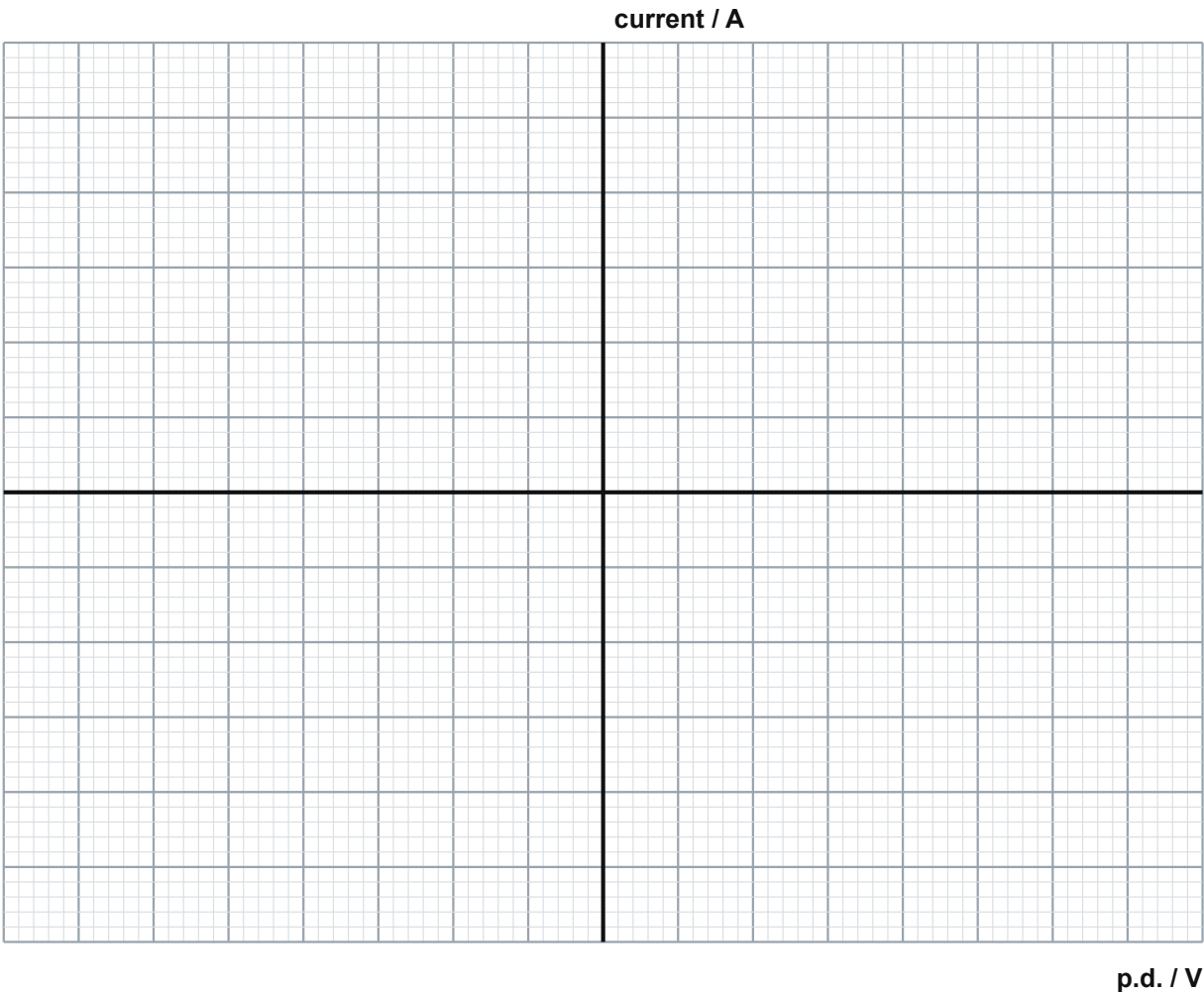
Q2 Calculate the resistance of the fixed resistor using one pair of your readings.

Equation
Substitute
Answer unit

Analysis Graphs and conclusions

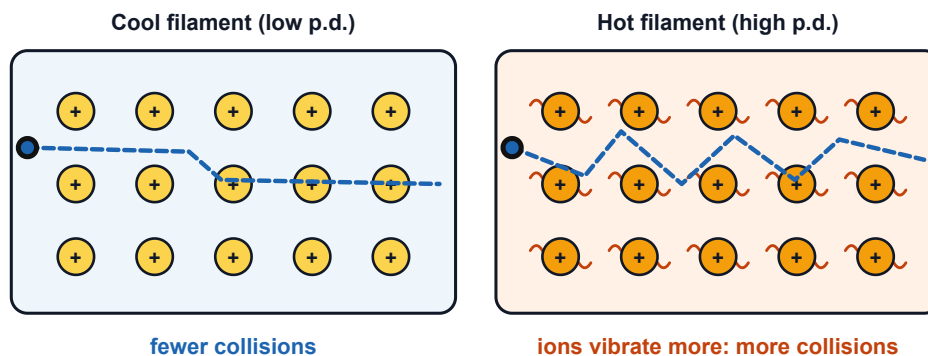
2 Filament lamp graph

Plot current against p.d. for the filament lamp. Add scales to both axes. Draw a smooth curve of best fit.



Q3 Use your readings to calculate the resistance of the lamp at your lowest and highest p.d. What happens to the resistance?

Analysis Graphs and conclusions



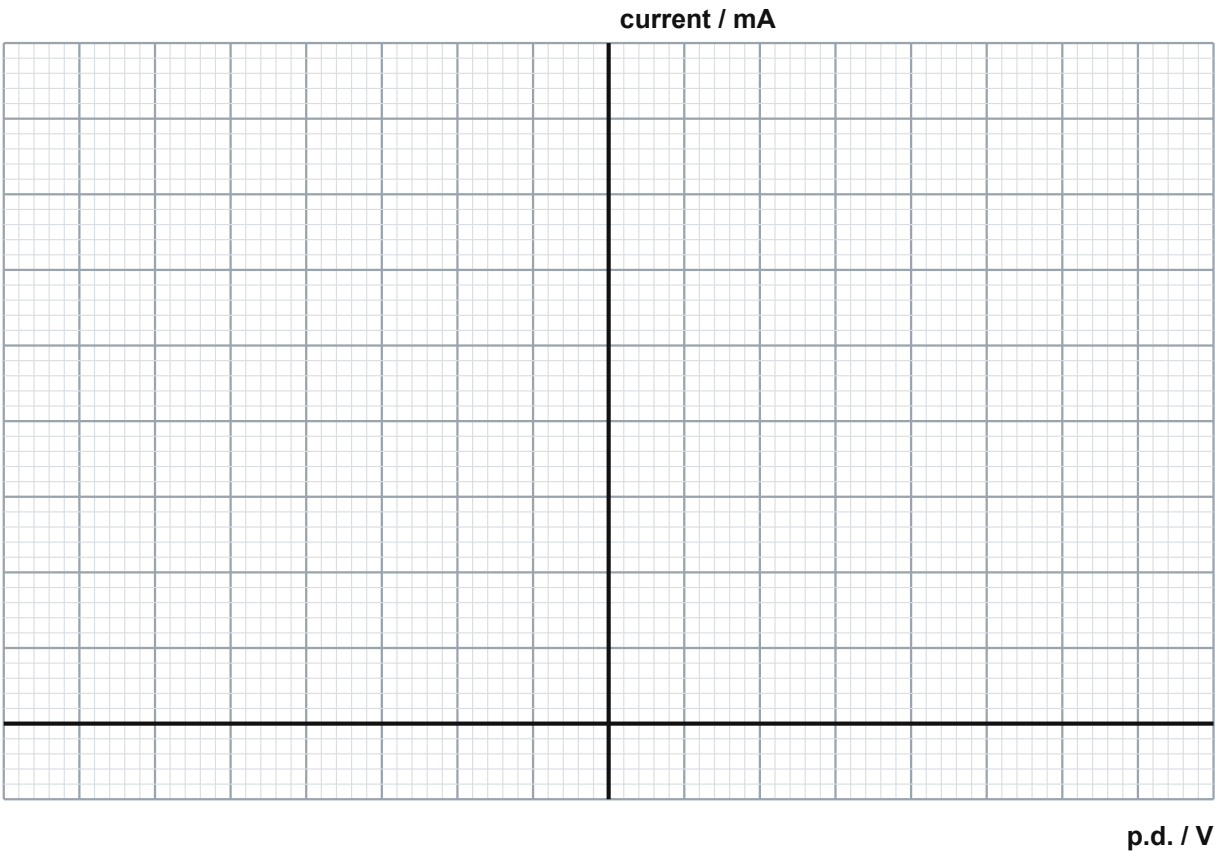
Electrons (blue path) collide with the metal ions (+). A hotter filament has vibrating ions, so there are more collisions and a higher resistance.

Q4 Explain why the resistance of the filament lamp changes. **[3 marks]**

Analysis Graphs and conclusions

3 Diode graph

Plot current against p.d. for the diode. Add scales to both axes. Draw a smooth curve of best fit.



Q5 Describe the I–V graph for the diode in the forward and reverse directions.

Q6 Explain why a protective resistor is connected in series with the diode.

Q7 Suggest why smaller steps in p.d. were used for the diode than for the other components.

Analysis Graphs and conclusions

4 Compare the three components

Complete the table using your three graphs.

Component	Shape of the I–V graph	As the p.d. increases, the resistance...	Ohmic?
Fixed resistor			
Filament lamp			
Diode			

Q8 A lamp has a p.d. of 4.5 V across it and a current of 0.26 A through it. Calculate its resistance.

Equation

Substitute

Answer

unit

Q9 Why were readings also taken with the supply reversed?

Q10 At the highest p.d. the fixed resistor felt warm. Suggest how this would affect the top end of its I–V graph.

Exam practice Plan the practical

I–V characteristics, Exam practice: A six-mark method question like the ones AQA asks about this practical. Write your answer without looking at the method.

AQA-STYLE **Q11** Describe a method to find how the current through a filament lamp depends on the p.d. across it. Include both positive and negative values of p.d. You may include a circuit diagram in your answer. **[6 marks]**

1 Check your answer

When you have finished, tick each point your method includes.

- | | |
|---|---|
| ☐ A circuit: lamp, ammeter and variable resistor in series | ☐ Record the current at each p.d.; stay below the rated p.d. |
| ☐ Voltmeter connected across the lamp only | ☐ Reverse the supply and record negative values |
| ☐ Change the p.d. with the variable resistor, in steps | ☐ Plot current (y) against p.d. (x); draw a smooth curve |